Project Report : CS 7643

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Abstract

Detecting bias in media has profound societal implications, yet it still remains a challenging task due to its nuanced nature and insufficient training data. Building upon the work of Spinde et al. [15], who introduced the BABE dataset and applied various attention-based models for the bias detection task, this study explores ways to further enhance performance of bias detection. We trained and optimized advanced attention-based models such as ERNIE, T5, and DeBERTa, and implemented data augmentation and distant pretraining to improve performance and tackle the limitations posed by the dataset's size. As a result, DeBERTa model combined with distant pretraining achieved a weighted macro f1 score of 0.812, outperforming Spinde et al. [15] with much less training data for distant pretraining. Our work underscores the importance of model selection, optimization and further training techniques in advancing the field of bias detection in media narratives.

1. Introduction

Media bias has a substantial influence on shaping public opinion and can greatly impact society. Detecting the language of bias can be challenging due to the nuanced nature of words and their context, especially when it comes to politics. Spinde et al. [15] introduced the BABE dataset (Bias Annotations By Experts) which attempts to address this with the collection of 3,700 sentences annotated by experts for bias. This data set was used with a variety of different advanced BERT-based models. In this paper, we will expand on the workflow of Spinde et al. [15] and use their methodology by experimenting with ERNIE, T5, and DeBERTa models. We will also consider additional methods to improve performance such as data augmentation and distant pre-training to determine if we can further improve the detection of media bias.

2. Background

Word-based sentence bias detection is a sub-area of media bias detection research. There are various known methods for this task. Statistical and machine learning approaches has been implemented for this task(Gummalam et al. [12]) and has reached up to 0.777 f1 score on the MBIC dataset(Spinde et al. [16]). However, because traditional models have limited ability to understand context, performance on detecting highly nuanced meaning is constrained.

More recently, Attention-based models have shown good performance, reaching up to 0.804 weighted macro f1 score with the RoBERTa model integrated with distant embeddings fine-tuning on the BABE dataset(Spinde et al. [15]). Even so, because sentence bias datasets are small in size(Spinde et al. [15]), there is a need for additional fine tuning methods to increase performance.

Nowadays, more context aware LLM has are appearing which are capable of few-shot learning(Brown et al [2]), and even zero-shot reasoning(Kojima et al. [13]. Since LLM are primarily used for text generation, sentence classification is not a focused research area. However, research on LLMs ability to detect bias in prompts has shown that LLMs indeed have the ability to detect bias in text(Nadeau et al. [14]). However, due to it's sheer size and expensive cost, LLMs are more suitable for generation related tasks(Brown et al [2]).

BABE (Bias Annotations by Experts, [11]) is a dataset designed for sentence based media bias detection. It consists of two sub-datasets: SG1, which contains 1,700 sentences from the MBIC dataset [16], gathered across 1,000 articles on 14 political topics between 2019 and 2020; and SG2, which includes 3700 manually extracted sentences from 14 US news platforms, covering 12 political topics between 2017 and 2020. For SG2, sentences were selected from articles of left-wing, center, and right-wing outlets.

The dataset was annotated by a group of carefully selected master's students with significant experience in media bias, using a detailed guideline and an iterative process with feedback. Annotations were only accepted with sufficient reasoning, and the final labels were determined

by a majority vote (SG1 - 8 annotators, SG2 - 5 annotators). Sentences without majority agreement were labeled as "no agreement". The dataset includes labels like "Factual", "Mixed", and "Opinionated", offering more detailed categorization. Figures 3 and 4 below visualize the dataset statistics.

3. Motivation

Bias in media narratives has profound societal implications, influencing public perception, political polarization, and even decision-making on critical global issues. Detecting and understanding such biases is essential for fostering a more informed and balanced public discourse. While most media outlets aim for objectivity, inherent framing choices can subtly shape opinions. If our project successfully improves bias detection accuracy, it could enhance tools for holding media outlets accountable and empower consumers to critically evaluate the information they consume.

Beyond academic interest, our project has practical applications. News aggregators, fact-checking organizations, and social platforms could leverage improved bias detection models to highlight diverse perspectives, mitigate the spread of one-sided narratives, and reduce the echo-chamber effects of algorithmic recommendations. Furthermore, refining these models can contribute to better AI alignment with human judgment, closing the gap between machine predictions and human perceptions of bias. In the long run, these advancements could play a pivotal role in building media literacy and promoting a more equitable information ecosystem.

4. Approach

We tackled bias detection by implementing three models: ERNIE, DeBERTa, and T5. For each model, we experimented with various sizes and hyperparameters to optimize performance. Performance is measured through the macro F1 score, which is defined as the arithmetic mean of the F1 scores of all classes. This is a suitable metric because it accounts for any potential class imbalance, such as that present in the SG1 variant. Additionally, we explored two enhancement strategies: data augmentation and distant pretraining. Data augmentation aimed to expand the BABE dataset, which, while high-quality, has a limited number of examples. We hypothesized that increasing the dataset size could improve model performance, especially given the scale of the models. Distant pretraining involved pre-training the models on a larger, lower-quality bias detection dataset to evaluate whether this could improve their performance on the BABE dataset by learning more generalized representations. Our work builds on [15] by introducing additional state-of-the-art models, exploring alternative datasets for the distant pretraining task, and applying data augmentation—an approach not explored in the reference paper.

For the code, we used the same pipeline in [11] in the *classification.ipynb* notebook, but altered the code to include our models. The T5 model required some changes in aspects such as the tokenization (custom tokenization was added to the pipeline to include the prompt along with the sentence in the input), the loss function (categorical cross entropy) and the prediction generation. To leverage the DeBERTa and ERNIE model and improve efficiency by streamlining processes, we also leveraged the transformer model selection with AutoModelForSequenceClassification which allows the code to simply switch between different models including different sizes of DeBERTa and ERNIE. We also used the Trainer API which automates different tasks such as logging, validation and computing metrics. The updates also include hyper parameter tuning, which allows for more functionality to various models and finding optimal models.

4.1. Challenges

4.1.1 Modeling

DeBERTa

The main challenge in using DeBERTa was striking the right balance between the size of the model and the adjustment of the parameter configurations, given the challenges with the dataset. DeBERTa, as a large transformer model, requires substantial training data to learn effectively without over-fitting. Pivoting and using the smallest version of the DeBERTa model increased training efficiency and allowed the model to perform better with the SG1 dataset. Limited data can lead to high variance and poor generalization, which drove the need for careful tuning and ensuring stability across each fold in training. Early performance was quite poor with more model parameters, resulting in a F1 score of 0.55. After adjusting the tokenization process, and using a smaller version of DeBERTa began to shift and improve performance. Carefully adjusting the model parameters such as learning rate, batch size, sequence length, and weight decay also improved perfromance. These considerations are critical to achieving competitive performance and getting the most out of this model.

ERNIE

The ERNIE model had similar challenges to the DeBERTa model. We selected the smallest ERNIE base model available [8], but still due to it's large model size(110 million parameters), the model was prone to over-fitting very early in the training process as shown in Figure 1. The only difference was that ERNIE showed good performance even without hyper parameter tuning. With k-fold cross validation(k=5) and initial hyper parameters, ERNIE showed 0.759 macro f1 score and 0.52 cross entropy loss on the SG1 dataset. However, tuning the hyper parameters had

little performance enhancement on the model. Tuning the batch size and learning did not have a significant performance change, and label smoothing also did not have an effect. Only weight decay had significant performance increase.

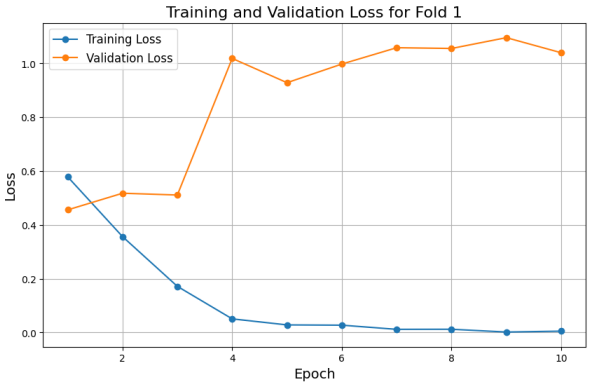


Figure 1. Example of ERNIE over-fitting

T5

The main challenge with the T5 architecture is that it is encoder-decoder, and thus does not support classification problems out of the box. We experimented with two T5 approaches: (1) T5 (encoder-decoder) and (2) T5 (encoder only + classification head). For version (1), we initially experimented with the basic T5 version [10], which is not prompt-trained, and thus performed poorly (Macro F1 score of 0.31). This version classified almost all sentences as Non-biased even with rigorous hyperparameter tuning. Then, we experimented with flan-T5 [9], which is prompt trained and thus we could give it prompts such as "Classify the news sentence after the colon as either '<s> Biased </s>' or '<s> Non-biased </s>'". This version performed better, but the F1 score still did not exceed 0.52. We noted that during training, the loss function decreased but this did not reflect in the F1 score, and thus it seemed like the loss metric (categorical cross entropy) was not a suitable fit for the task. Additionally, both T5 and FlanT5 had the downside of generating classes that were not "Biased" or "Non-biased", such as "Bipartisan" and "Bipolar". Despite multiple attempts with prompt engineering, this issue could not be avoided and thus we relied on post-processing to map the model output to one of the 2 classes. The aforementioned challenges led us to our next experiment, which uses only the encoder of the FlanT5 model, and applies a classification head over the encoded representation. This version performed much better, even with the same prompt and hyperparameters. Further hyperparameter tuning was then performed on the flant5-small and the flant5-base versions of the model until a satisfactory performance was reached.

4.1.2 Data Augmentation

To further increase the performance, we tried data augmentation. There were two strategies, paraphrasing the original data and generating new data. With a T5 model fine-tuned for paraphrasing we created 200 paraphrased sentences from the original dataset. On the other hand, with the chatGPT-4o model we used various prompts to generate 200 new sentences. We further optimized the augmentation process by finding the optimal temperature value for the T5 model, and finding the optimal prompt for the chatGPT model. Optimizing data augmentation based on various hypotheses showed positive performance improvement.

Hyper parameter and prompt optimization were the main task of data augmentation. The temperature hyper parameter controls the randomness of the T5 model output, which means that increasing the value allows the model to explore more lower-probability outputs and vice versa. Due to this, the temperature value had significant effect on data augmentation performance. Additionally, we experimented with various prompt versions to find the prompt which led to the most performance-enhancing data augmentation.

4.1.3 Distant Pretraining

In their distant pretraining experiment, Spinde et al. [15] propose that pretraining models on a large but noisy bias detection dataset can enhance performance on the BABE dataset. The dataset they used was constructed by scraping 128,746 headlines containing specific keywords from various news outlets. Each sentence was then labeled based on the political leaning of its source, as determined by the outlet's position on the Media Bias Chart [1]. However, this approach introduces a significant challenge: the inherent noise in the dataset, as not every headline from a biased news source is necessarily biased. While such noise is often acceptable in pretraining tasks, it can limit the effectiveness of the pretraining process and hinder model performance.

In contrast, we sought to evaluate the impact of a more accurately labeled dataset. For this purpose, we selected the Political Social Media Posts dataset [7], which comprises 5,000 social media messages from U.S. politicians, annotated by non-experts as either "neutral" or "partisan." While this dataset is much smaller in size, its higher annotation quality reduces noise and addresses one of the challenges posed by the larger dataset. However, this introduces a new challenge: the trade-off between dataset size and label reliability. The smaller scale of this dataset raises the question of whether its higher quality can compensate for its reduced size in pretraining. This provides a unique opportunity to test whether high-quality data can overcome the limitations of smaller datasets, particularly in the context of bias detection.

4.1.4 Subtle Bias

The models had a bias towards biased sentences that contained explicit words and unbiased sentences that contained factual content. After training models, we checked the prediction results on the validation data. We found that the biased sentences which the models were predicting correctly included explicit expressions. For example *"Nevertheless, the media jihad continues, and that includes NBC shamelessly running a debunked and deceptive smear story that had been reported on and debunked all the way back in May."* contains explicit words such as "shamelessly", "deceptive", and "smear". On the other hand, biased sentences with the models predicted as unbiased did not contain any explicit or dramatic expressions, and only had words that are biased in a certain context - which the model could not know without further information. For instance, *"Four states will hold their primary contests on Tuesday night, despite Georgia and Louisiana each postponing their elections previously scheduled for a few weeks later over concerns of the spreading Wuhan virus."* is labeled biased because of the words "Wuhan" and "Virus", which were controversial words when the article was published. Full results are available in our github repository [5].

To confirm our finding, we analyzed the label opinion values of the results. As we can see in Figure 2, more than half of the biased sentences that were predicted correctly were labeled `Opinionated`, while the biased sentences which the model got wrong was mostly labeled as `Partially Opinionated`. Furthermore, unbiased sentences which the model got right was mostly labeled `Factual`, while the counterpart had varying labels. This shows that the model had difficulties predicting biased sentences without explicit vocabulary which expressed the writer's opinion, and unbiased sentences which lacked factual content. The jupyter notebook code used for analysis is available here [6].

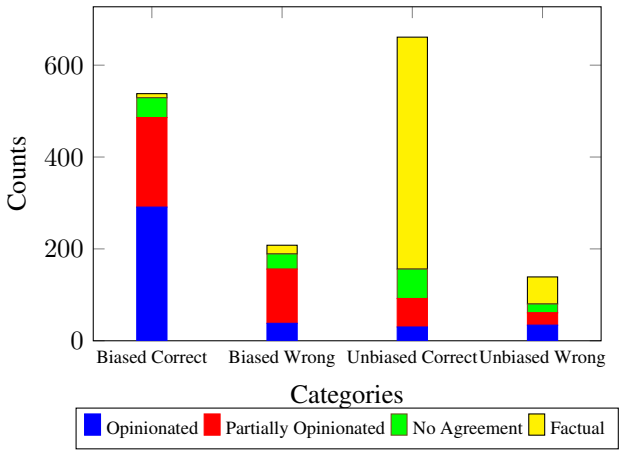


Figure 2. Distribution of label opinions.

Model	Macro F1	
	SG1	SG2
Baseline **	0.68	0.72
BERT **	0.750	0.790
BERT + distant (news headlines) **	0.778	0.804
RoBERTa + distant (news headlines)**	0.775	0.799
ERNIE	0.781	0.796
DeBERTa	0.806	0.81
T5 (encoder + decoder)	0.525	0.510
T5 (encoder + classification)	0.786	0.804
BERT + data augmentation	0.771	-
ERNIE + data augmentation	0.785	-
DeBERTa + data augmentation	**0.812**	-
DeBERTa + distant (social media)	**0.812**	**0.812**
DeBERTa + distant (news headlines)	0.810	0.804

Table 1. Stratified 5 fold cross-validation results. The models marked with ** are built by [15].

5. Experiments and Results

The goal of our experiments is to enhance model performance for bias detection. We utilized techniques like data augmentation, distant pretraining, and hyper parameter tuning to optimize F1 scores and improve model generalization. Data augmentation addressed dataset size limitations with paraphrasing (using a T5 model) and sentence generation (using ChatGPT). We also explored distant supervision with social media data to assess its impact on performance. Hyper parameter tuning was conducted to identify optimal settings for each model. The following results summarize the effectiveness of these approaches.

5.1. ERNIE

We planned to tune each hyper parameter separately and then try different combinations of the hyper parameters which showed positive results. However, as we can see in Table 5, only weight decay resulted in both increase in macro f1 and decrease of loss. Thus, only weight decay was applied to the model. The results of the tuning experiments are shown in Table 5

5.2. T5

5.2.1 Encoder + decoder

In this experiment, we ran the full encoder-decoder T5 model on the dataset. The task was modified to become a text-to-text task as follows: the input was a prompt, specifically "Classify the news sentence after the colon as either '<s> Biased </s>' or '<s> Non-biased </s>:'". Don't output any other word.:". The output of the model, which ideally was either `Biased` or `Non-biased` but also included other random words sometimes, was mapped back

to a class of 0 (Non-biased) and 1 (Biased), and then evaluation would proceed as normal by comparing the predicted class against the ground truth. An initial version used the normal T5 model in [10], and then it was replaced by FlanT5 which is prompt-trained. We experimented with a range of short and long prompts, ranging from having the sentence only with no additional text, to a long prompt that mimics the guidelines given to the BABE dataset annotators, and the best one was identified to be the one specified at the beginning of this paragraph. As seen in Table 1, the results of this version of the T5 model were subpar and did not even beat the baseline, looking at examples of misclassifications showed that the model was misclassifying even easy examples, and was outputting words that do not belong to either classes. Since this model did not seem to be learning effectively, we decided to switch to another architecture that only uses the encoder part of the architecture.

5.2.2 Encoder only + classification head

The objective of this experiment is to evaluate the performance of a T5 encoder-only architecture with a classification head on a binary classification task. The goal is to determine the effectiveness of the T5 encoder in extracting meaningful representations from textual data and its capability to support downstream classification tasks. The classification head is a dense layer with 2 neurons followed by a softmax activation. In the first iteration of this model, the model was suboptimal (F1 score of 0.5 for SG1). However, tuning hyperparameters such as the number of epochs, batch size, learning rate, and input prompt significantly improved the performance, resulting in an F1 score of 0.71 for the SG1 dataset. All experiments in this phase utilized the flan-t5-small version of the model. Some tuning experiments on the SG1 dataset and their results are documented in Table 6. Our tuning approach always involved changing one hyperparameter at a time, and observing the effect on the F1 score.

Subsequently, we experimented with the flan-t5-base version, which yielded further performance gains, as shown in Table 1. This improvement is consistent with expectations, as the increased parameter count in the flan-t5-base model enabled it to capture more nuanced representations of the input sentences.

5.3. DeBERTa

The experiment evaluated on the performance of DeBERTa was also based on the classification tasks of identifying whether a sentence was “Biased” or “Non-biased”. The primary objective was to assess the model’s ability to generalize on the BABE dataset and determine the impact of hyper parameter tuning and scaling. A variety of adjustments were made during the experimental phase. This included testing a variety of different versions of DeBERTa including ‘microsoft/deberta-v3-small’, ‘microsoft/deberta-v3-base’ and ‘microsoft/deberta-v3-large’. There were also adjustments to the tokenization process to determine appropriate length of sequence tokens. The initial results yielded suboptimal results, with inconsistent F1 scores across folds. The model struggled to differentiate biased and non-biased sentences with default configurations. The large DeBERTa model struggled to learn efficiently with the smaller dataset SG1 dataset. Adjusting hyper parameters did not seem to show improvement. The smallest version of the model had shown the most promising results with limited need for additional tuning. Adjustments had to be made including reducing the learning rate to 1e-4, decay to 0.01, using a batch size of 16 for training and testing, and truncating the tokens to 150. Reference the different parameters and F1 score results here: 7. Overall, the DeBERTa model showed strong performance with the classification task, displaying F1 scores around 0.81 for the SG2 dataset. Tuning the smaller model demonstrated importance when using these datasets.

5.4. Data Augmentation

This experiment was conducted on the SG1 dataset, which has only 1700 observations. Performance experiments were performed with the ERNIE and DeBERTa model, and the best data augmentations were tested on the BERT model to see whether the effect was limited to the 2 models only.

For the experiments, hyper parameters were fixed and 200 sentences were created from each augmentation method [4] and added to the training set (not the validation set).

5.4.1 Paraphrase

We thought that models could be more familiar with certain words or phrases, so by providing them with various expressions we can increase the chances of the model to understand the nuance of the sentence.

We used the T5 model fine-tuned for paraphrasing, and experimented with two temperature values, 1.3 and 1.5. Temperature value of 1.3 resulted in sentences which were more similar to the original sentence and used almost the same words. Temperature value of 1.5 resulted in more varying vocabulary, and the sentences sometimes did not make sense.

5.4.2 Generation

We believed that paraphrased sentences could make the model more susceptible to over-fitting since it does not provide data with new meaning. So therefore, by generating new sentences from scratch, we thought that we could provide more cases for the model to learn from. We used

ChatGPT-4o with varying prompts [3]. There were five versions, zero-shot(V1), few-shot(v2), few-shot with instructions(v3), instructions with clear biased/unbiased examples(v4), and instructions with examples which were in the gray area(v5).

5.4.3 Results

The T5(temperature=1.5) and ChatGPT prompt version 3 had the best results. The result was ranked by the sum of the increase of macro f1 and the decrease of loss.

Type	Value	Macro F1	Loss
T5	1.3	0.755	0.517
T5	1.5	**0.781**	**0.499**
ChatGPT	v1	0.769	0.507
ChatGPT	v2	0.770	0.510
ChatGPT	v3	**0.785**	**0.497**
ChatGPT	v4	0.767	0.520
ChatGPT	v5	0.777	0.508

Table 2. Data Augmentation results on ERNIE

With data augmentation, we were able to reach 0.785 weighted macro f1 score with the ERNIE model. Furthermore, by applying data augmentation to our best performing DeBERTa model, we were able to reach 0.812, tying the best score for SG1 with deBERTa with distant pretraining.

Afterwards, the augmented data was further tested on the BERT model to see whether there were same positive effects. Besides the fact that ChatGPT version 3 data resulted in decrease in performance, other results aligned with the ERNIE results.

Type	Value	Macro F1	Loss
baseline	-	0.750	0.524
T5	1.5	**0.761**	**0.496**
ChatGPT	v3	0.737	0.5321
ChatGPT	v4	**0.771**	**0.496**
ChatGPT	v5	0.756	0.516

Table 3. Data Augmentation results on BERT

5.5. Distant Pretraining

The results in Table 1 demonstrate that distant pretraining on the political social media dataset enhances DeBERTa's performance by 0.006 on the SG1 dataset and 0.002 on the SG2 dataset. While these improvements are modest, they are particularly notable when compared to the distant pretraining conducted on the larger news headlines dataset, which included 25 times more instances but yielded a smaller improvement (0.004 on the SG1 dataset and worse performance on the SG2 dataset). This suggests that, at least for this task, pretraining on a higher-quality dataset may be more effective than using a larger dataset. The relatively small gain also indicates that the DeBERTa model has already reached its performance boundary on the more straightforward bias instances, with the remaining difficult examples, as discussed in section 4.1.4, being challenging even for humans to classify.

5.6. Summary

We were successful in our goal to enhance model performance for bias detection. We utilized techniques like data augmentation, distant pre-training, and hyper parameter tuning to optimize F1 scores and improve model generalization. All three models (DeBERTa, ERNIE, and T5) surpassed the best-performing model from prior work, RoBERTa, even without distant pre-training. Data augmentation consistently improved model performance by diversifying the training data and reducing over-fitting, with the best configurations leading improvement in F1. Similarly, distant supervision further enhanced performance by utilizing a related task from a social media bias dataset to provide additional training signals. Overall, the combination of these strategies resulted in DeBERTa achieving the highest Macro F1 score of 0.812, showcasing the robustness and effectiveness. We were able to leverage advanced techniques and models to outperform previous research.

6. Work Division

HyunDuk Seo optimized the hyperparameters for the ERNIE model and applied various data augmentation 5.4 methodologies to enhance performance. Yasmeen Abdelhafeez focused on training T5 model variants and conducted a social media pretraining experiment. Cole Ofsthun trained the DeBERTa model, developed a baseline model for comparison, and created dataset visualizations.

4.

Student Name	Contributed Aspects	Details
HyunDuk Seo	ERNIE model and data augmentation ??	Optimized the hyperparameter for the ERNIE model, and augmented data using various methodologies to further improve performance.
Yasmeen Abdelhafeez	T5 model and distant pretraining	Trained the T5 model variants and implemented the social media pretraining experiment
Cole Ofsthun	Baseline and DeBERTa models	Trained DeBERTa model and created baseline model for comparison and dataset visualization

Table 4. Contributions of team members.

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A. Appendix

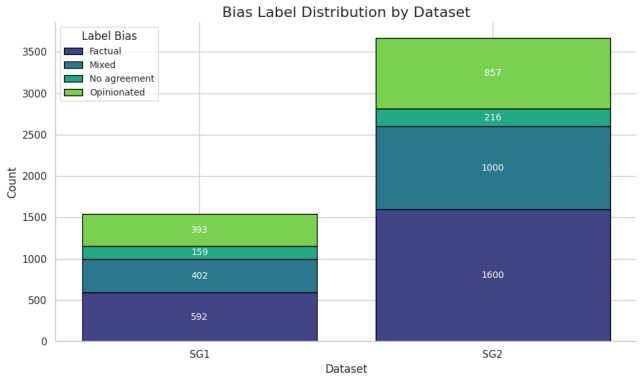


Figure 3. Label Opinion distribution

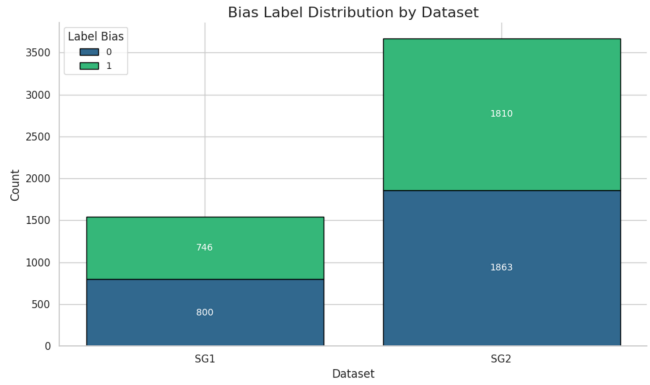


Figure 4. Bias Label Distribution

Parameter	Value	Macro F1	CV Loss
Baseline	-	0.759	0.525
Weight Decay	0.005	0.773	0.486
Weight Decay	0.01	0.774	0.486
Weight Decay	0.5	**0.781**	**0.487**
Label Smoothing	True	0.772	0.537
Batch Size	8	0.754	0.547
Batch Size	16	0.772	0.539
Batch Size	128	0.776	0.533
Batch Size	256	0.740	0.578
Learning Rate	1e-5	0.757	0.542
Learning Rate	8e-5	0.765	0.540
Learning Rate	1e-4	0.756	0.516

Table 5. ERNIE hyperparameter Tuning Results on SG1

n_epochs	batch_size	lr	prompt	Macro F1
2	8	1e-5	v1	0.5
5	8	1e-5	v1	0.66
5	8	2e-5	v1	0.7
5	16	2e-5	v1	0.71
5	16	2e-5	v2	0.67
5	16	2e-5	v3	0.63
5	16	2e-5	v4	0.710
5	**16**	**2e-5**	**v5**	**0.714**
10	16	2e-5	v5	0.7

Table 6. T5 hyperparameter Tuning Results

Model Size	Learning Rate	Decay	Macro F1
Small	5e-5	0.1	0.806
Small	1e-4	0.1	0.807
Small	5e-5	0.01	0.807
Small	1e-4	0.01	0.810
Base	5e-5	0.1	0.797
Base	1e-4	0.1	0.599
Base	5e-5	0.01	0.792
Base	1e-4	0.01	0.596
Large	5e-5	0.1	0.617
Large	1e-4	0.1	0.685
Large	5e-5	0.01	0.632
Large	1e-4	0.01	0.612

Table 7. DeBERTa Hyperparameter Tuning Results